

Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Code:

```
import cv2

import matplotlib.pyplot as plt

from google.colab import files

# Upload image

uploaded = files.upload()

# Read image

img = cv2.imread(list(uploaded.keys())[0])

# Plot histogram for color levels

colors = ('b', 'g', 'r')

for i, col in enumerate(colors):

    hist = cv2.calcHist([img], [i], None, [256], [0, 256])

    plt.plot(hist, color=col)

plt.title("Color Histogram")

plt.xlabel("Pixel Value")

plt.ylabel("Frequency")

plt.show()
```

Output:



